# An Analysis of Principal component analysis

*Principal component analysis*

## Principal component analysis -- Part 1

in such a way that the individual variables t 1 , … , t l {\displaystyle t_{1},\dots ,t_{l}} of t considered over the data set successively inherit the maximum possible variance from X, with each coefficient vector w constrained to be a unit vector. The above may equivalently be written in matrix form as

## Principal component analysis -- Part 2

E ( α ) = 1 1 − α 2 ∑ i = 1 n ( α p i − q i ) 2 {\displaystyle E(\alpha )={\frac {1}{1-\alpha ^{2}}}\sum _{i=1}^{n}(\alpha p_{i}-q_{i})^{2}} ; such that setting the derivative of the error function to zero ( E ′ ( α ) = 0 ) {\displaystyle (E'(\alpha )=0)} yields: α = 1 2 ( − λ ± λ 2 + 4 ) {\displaystyle \alpha ={\frac {1}{2}}\left(-\lambda \pm {\sqrt {\lambda ^{2}+4}}\right)} where λ = p ⋅ p − q ⋅ q p ⋅ q {\displaystyle \lambda ={\frac {p\cdot p-q\cdot q}{p\cdot q}}} .

## Principal component analysis -- Part 3

Non-negative matrix factorization Non-negative matrix factorization (NMF) is a dimension reduction method where only non-negative elements in the matrices are used, which is therefore a promising method in astronomy, in the sense that astrophysical signals are non-negative. The PCA components are orthogonal to each other, while the NMF components are all non-negative and therefore constructs a non-orthogonal basis. In PCA, the contribution of each component is ranked based on the magnitude of its corresponding eigenvalue, which is equivalent to the fractional residual variance (FRV) in analyzing empirical data. For NMF, its components are ranked based only on the empirical FRV curves. The residual fractional eigenvalue plots, that is, 1 − ∑ i = 1 k λ i / ∑ j = 1 n λ j {\displaystyle 1-\sum _{i=1}^{k}\lambda _{i}{\Big /}\sum _{j=1}^{n}\lambda _{j}} as a function of component number k {\displaystyle k} given a total of n {\displaystyle n} components, for PCA have a flat plateau, where no data are captured to remove the quasi-static noise, then the curves drop quickly as an indication of over-fitting (random noise). The FRV curves for NMF is decreasing continuously when the NMF components are constructed sequentially, indicating the continuous capturing of quasi-static noise; then converge to higher levels than PCA, indicating the less over-fitting property of NMF.

## Principal component analysis -- Part 4

Discriminant analysis of principal components Discriminant analysis of principal components (DAPC) is a multivariate method used to identify and describe clusters of genetically related individuals. Genetic variation is partitioned into two components: variation between groups and within groups, and it maximizes the former. Linear discriminants are linear combinations of alleles which best separate the clusters. Alleles that most contribute to this discrimination are therefore those that are the most markedly different across groups. The contributions of alleles to the groupings identified by DAPC can allow identifying regions of the genome driving the genetic divergence among groups In DAPC, data are first transformed using a principal components analysis (PCA) and subsequently clusters are identified using discriminant analysis (DA). A DAPC can be realized on R using the package Adegenet. (more info: adegenet on the web)

## Principal component analysis -- Part 5

Population genetics In 1978 Cavalli-Sforza and others pioneered the use of principal components analysis (PCA) to summarise data on variation in human gene frequencies across regions. The components showed distinctive patterns, including gradients and sinusoidal waves. They interpreted these patterns as resulting from specific ancient migration events. Since then, PCA has been ubiquitous in population genetics, with thousands of papers using PCA as a display mechanism. Genetics varies largely according to proximity, so the first two principal components actually show spatial distribution and may be used to map the relative geographical location of different population groups, thereby showing individuals who have wandered from their original locations. PCA in genetics has been technically controversial, in that the technique has been performed on discrete non-normal variables and often on binary allele markers. The lack of any measures of standard error in PCA are also an impediment to more consistent usage. In August 2022, the molecular biologist Eran Elhaik published a theoretical paper in Scientific Reports analyzing 12 PCA applications. He concluded that it was easy to manipulate the method, which, in his view, generated results that were 'erroneous, contradictory, and absurd.' Specifically, he argued, the results achieved in population genetics were characterized by cherry-picking and circular reasoning.

## Principal component analysis -- Part 6

Software/source code ALGLIB – a C++ and C# library that implements PCA and truncated PCA Analytica – The built-in EigenDecomp function computes principal components. ELKI – includes PCA for projection, including robust variants of PCA, as well as PCA-based clustering algorithms. Gretl – principal component analysis can be performed either via the pca command or via the princomp() function. Julia – Supports PCA with the pca function in the MultivariateStats package. KNIME – A java based nodal arranging software for Analysis, in this the nodes called PCA, PCA compute, PCA Apply, PCA inverse make it easily. Maple (software) – The PCA command is used to perform a principal component analysis on a set of data. Mathematica – Implements principal component analysis with the PrincipalComponents command using both covariance and correlation methods. MathPHP – PHP mathematics library with support for PCA. MATLAB – The SVD function is part of the basic system. In the Statistics Toolbox, the functions princomp and pca (R2012b) give the principal components, while the function pcares gives the residuals and reconstructed matrix for a low-rank PCA approximation. Matplotlib – a Python library that has a PCA package in the .mlab module. mlpack – Provides an implementation of principal component analysis in C++. mrmath – A high performance math library for Delphi and FreePascal can perform PCA; including robust variants. NAG Library – Principal components analysis is implemented via the g03aa routine (available in both the Fortran versions of the Library). NMath – Proprietary numerical library containing PCA for the .NET Framework. GNU Octave – Free software computational environment mostly compatible with MATLAB, the function princomp gives the principal component. OpenCV Oracle Database 12c – Implemented via DBMS_DATA_MINING.SVDS_SCORING_MODE by specifying setting value SVDS_SCORING_PCA. Orange (software) – Integrates PCA in its visual programming environment. PCA displays a scree plot (degree of explained variance) where user can interactively select the number of principal components. Origin – Contains PCA in its Pro version. Qlucore – Commercial software for analyzing multivariate data with instant response using PCA. R – Free statistical package, the functions princomp and prcomp can be used for principal component analysis; prcomp uses singular value decomposition which generally gives better numerical accuracy. Some packages that implement PCA in R, include, but are not limited to: ade4, vegan, ExPosition, dimRed, and FactoMineR. SAS – Proprietary software; for example, see. scikit-learn – Python library for machine learning which contains PCA, Probabilistic PCA, Kernel PCA, Sparse PCA and other techniques in the decomposition module. Scilab – Free and open-source, cross-platform numerical computational package, the function princomp computes principal component analysis, the function pca computes principal component analysis with standardized variables. SPSS – Proprietary software most commonly used by social scientists for PCA, factor analysis and associated cluster analysis. Weka – Java library for machine learning which contains modules for computing principal components.